Journal of Hydrology 645 (2024) 132231

Contents lists available at ScienceDirect

Journal of Hydrology

journal homepage: www.elsevier.com/locate/jhydrol

Research papers

Relationship between root characteristics and saturated hydraulic conductivity in a grassed clayey soil

Ling-Xin Cui [a], Qing Cheng [a,*], Pui San So [b], Chao-Sheng Tang [a], Ben-Gang Tian [a], Cong-Ying Li [a]

[a] *School of Earth Sciences and Engineering, Nanjing University, 163 Xianlin Road, Nanjing 210023, China*
[b] *Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong Special Administrative Region*

A R T I C L E I N F O

Keywords:
Saturated hydraulic conductivity
Dry density
Planting density
Root parameters

A B S T R A C T

Soil saturated hydraulic conductivity plays a crucial role in the fields of hydrology, geotechnical and geological engineering. This study investigated the relationship between root characteristics and soil saturated hydraulic conductivity in a grassed soil. The saturated hydraulic conductivity of soil specimens with varying soil dry densities and planting densities were experimentally determined. Root parameters of each specimen were measured, and the relationship between root parameters and soil saturated hydraulic conductivity was determined through the stepwise multiple regression analysis. Experimental results show that both soil dry density and planting density significantly influence root growth and the saturated hydraulic conductivity of the grassed soil. Root length ratio (>4 cm), root length, and root weight density exhibit a positive correlation with planting density, while root length density and root length ratio (>4 cm) are negatively correlated with dry density. Higher dry density leads to lower soil saturated hydraulic conductivity and the permeable effects become more pronounced with increased planting density. When planting density is higher, roots create more preferential flow channels in the soil, resulting in increased soil saturated hydraulic conductivity. Root length density exerts the most significant effect on soil saturated hydraulic conductivity ($|r| = 0.82$, $p < 0.01$), followed by root length ratio (>4 cm) and root weight density. This study provides insights into the relationship between root parameters and soil saturated hydraulic conductivity, and identifies key root parameters that govern the saturated hydraulic conductivity of soil. These findings hold significant implications for enhancing the understanding of slope protection using vegetation.

1. Introduction

Soil saturated hydraulic conductivity is a fundamental property that governs various hydrological and geotechnical processes. From a hydrological perspective, soil saturated hydraulic conductivity plays a pivotal role in governing surface water cycle, facilitating groundwater recharge and impacting agricultural production (Pandey and Pandey, 1982; Liu et al., 2019; Ledford et al., 2020). It is also crucial in geotechnical or geological engineering to evaluate the performance of earthen structures, soil erosion, slope stability and settlement of foundation (Fragaszy et al., 2011; Cheng et al., 2021; Zhang et al., 2023; Wu et al., 2024). Many researchers have extensively investigated soil saturated hydraulic conductivity, recognizing soil texture and density as pivotal factors (Hwang et al., 2004; Ding et al., 2016; Ben et al., 2009). Generally, large clay content or large compaction density reduces soil porosity and pore connectivity, resulting in decreased soil saturated hydraulic conductivity, and vice versa (Wang et al., 1984; Ben et al., 2009). Moreover, several studies have demonstrated that the presence of organic matter in soil generally enhances soil saturated hydraulic conductivity. Organic matter decomposition results in the formation of smaller particles and micro-aggregates, thereby increasing soil porosity and facilitating saturated hydraulic conductivity (Usharani et al., 2019). Furthermore, environmental conditions, particularly extreme weather events resulting from climate change (such as heavy precipitation and droughts), significantly influence soil saturated hydraulic conductivity. Rainfall increases soil water content, consequently raising the saturated hydraulic conductivity (Chen et al., 2013; Ran et al., 2018). Prolonged droughts can cause soil desiccation and cracking, reducing water retention capacity while simultaneously enhancing saturated hydraulic conductivity (Bordoloi et al., 2020; Tang et al., 2021).

* Corresponding author.
E-mail addresses: 522022290004@smail.nju.edu.cn (L.-X. Cui), chengqing@nju.edu.cn (Q. Cheng), cepsso@ust.hk (P.S. So), tangchaosheng@nju.edu.cn (C.-S. Tang), tianbengang@smail.nju.edu.cn (B.-G. Tian), 522022290011@smail.nju.edu.cn (C.-Y. Li).

https://doi.org/10.1016/j.jhydrol.2024.132231
Received 27 May 2024; Received in revised form 5 October 2024; Accepted 8 October 2024
Available online 22 October 2024

In recent years, vegetation has been widely acknowledged as an effective natural-based solution mitigating various engineering problems and disasters (Kim et al., 2017; Gonzalez and Mickovski, 2017; Fan et al., 2022; Cheng et al., 2024). Plant roots reinforce soil and enhance its strength (De et al., 2008; Stokes et al., 2009; Ng et al., 2013). Additionally, transpiration by plants increases soil matric suction and, hence, soil shear strength (Leung et al., 2015a; Ni et al., 2017). When the effects of plants on soil hydraulic behavior are concerned, most studies suggest that plant roots occupy a part of soil pores, potentially reducing soil saturated hydraulic conductivity (Colombi et al., 2018; Chen et al., 2021; Guo et al., 2022; Leung et al., 2023). Other studies also concluded that the interactions between plant roots and soil particles can increase saturated hydraulic conductivity by creating preferential flow channels and enhancing connectivity among soil pores (Shao et al., 2017; Banwart et al., 2019; Lu et al., 2020; Shi et al., 2021). Although root parameters like length, surface and volume density were quantified in these studies to examine their effects on soil pores, their linkage with saturated hydraulic conductivity was only assessed qualitatively and indirectly. Overall, the impact of plant roots on soil saturated hydraulic conductivity remains unclear. It is necessary to consider the root-soil interactions more comprehensively and establish quantitative correlations between root parameters and saturated hydraulic conductivity, thereby further improving the performance of vegetated earthen structures.

The principal objective of this study is to investigate the impact of ryegrass roots on soil saturated hydraulic conductivity. A series of saturated hydraulic conductivity tests were carried out on rooted and bare soil specimens with four soil dry densities of 1.3, 1.4, 1.5, and 1.6 Mg/m^3. Three planting densities of 24, 48, and 72 g/m^2 were considered. Moreover, root parameters including root length density, root weight density and root length ratio were measured. After that, the quantitative relationship between these root parameters and soil saturated hydraulic conductivity were established. This research can offer valuable insights into the influence of plant roots on soil saturated hydraulic conductivity, elucidate the relationship between plant root parameters and soil saturated hydraulic conductivity, and determine root parameters that affect saturated hydraulic conductivity. These findings can provide guidance for engineering applications involving plants, such as green slope engineering and ecological restoration.

2. Materials and methods

2.1. Soil and plant species

The soil utilized in this study is the Xiashu soil, which is widely distributed in the middle and lower reaches of the Yangtze River. The soil composition comprises 2 % sand, 76 % silt, and 22 % clay.

The maximum dry density and optimum gravimetric water content were determined based on the dynamic compaction curve obtained by the standard Proctor compaction test (ASTM D698-12, 2012). With a given compaction effort, the dry density of soil increases first and then decreases with the increase of water content. The water content corresponding to the peak point is called the optimum gravimetric water content and the density corresponding to the peak point is called the maximum dry density. The liquid and plastic limits are 36.5 % and 19.5 %, respectively. Index properties of the test soil are shown in Table 1. According to ASTM D-2481 (2017), it is classified as a lean clay (CL).

Ryegrass (*Lolium perenne L.*), an herbaceous plant known for its rapid growth and drought resilience, was selected. This plant species is widely used in China to control soil erosion, especially for the Yangtze River Basin (Zhou and Shangguan, 2007).

2.2. Specimen preparation

The experiment consists of two groups: a rooted soil test group labeled with an "R" prefix and a bare soil group labeled with a "B" prefix. Each group encompasses four soil dry densities (1.3, 1.4, 1.5, and 1.6 Mg/m^3). Within the rooted soil test group, three planting densities (24, 48, and 72 g/m^2) are considered to prepare specimens at a given soil dry density. Planting density refers to the mass of ryegrass seeds sown per unit area of soil (Cheng et al., 2024). The test conditions are summarized in Table 2. Three replicates are prepared for each test condition.

The soil was first oven-dried and passed through a 2-mm sieve. Subsequently, the soil was thoroughly mixed with de-aired water to reach its optimum gravimetric water content (16.5 %). The soil mixture was then sealed in a bag for 48 h to ensure homogenous moisture distribution. Static compaction is adopted to prepare the soil specimens at four desired dry densities. The dimension of the specimens was 240 mm in length, 150 mm in width and 50 mm in height. Each specimen was statically compacted in five layers. For the ryegrass-planted soil specimen, after the compaction, pre-germinated grass seeds were evenly sowed on the soil surface using hydroponics at planting densities of 24 g/m^2, 48 g/m^2, and 72 g/m^2 respectively. The purpose of utilizing hydroponics is to facilitate the pre-germination of grass seeds prior to sowing, thereby enhancing their survival rate. Besides, seeds were not sowed for the bare soil specimens. The bare and rooted soil specimens were consistently watered once every day under controlled environmental conditions (T = 25 ± 1 ℃, RH = 60 ± 2 %), a plant growth lamp was positioned 30 cm above the surface of the ryegrass specimens, with the light intensity maintained at approximately 5000 lx. After plant growth, all grass blades on the surface of the rooted soil specimens were trimmed. Subsequently, a cutting ring with dimensions of 61.8 mm in diameter and 40 mm in height was employed to extract specimens from the mold. Prior to sampling, a uniform layer of grease was applied to the inner wall of the cutting ring, the resistance during insertion into the soil is reduced and specimen removal is facilitated. Before the test began, the specimens were immersed in water for 48 h for saturation. Filter paper and porous stone were placed on the top and bottom surfaces of the specimen. A vacuum pump was connected to the container to facilitate

Table 1
Index properties of Xiashu soil.

Property	Value
Specific gravity, *Gs*	2.73
Liquid limit, w_l (%)	36.5
Plastic limit, w_p (%)	19.5
Plasticity index, I_p (%)	17.0
Shrinkage limit, w_{sl} (%)	10.5
Optimum gravimetric water content, w_{op} (%)	16.5
Maximum dry density (Mg/m^3)	1.7
Sand content (%)	2
Slit content (%)	76
Clay content (%)	22

Table 2
Details of the saturated hydraulic conductivity tests.

Test ID	Planting density (g/m^2)	Dry density of soil (Mg/m^3)
BD13	0	1.3
BD14		1.4
BD15		1.5
BD16		1.6
R24D13	24	1.3
R24D14		1.4
R24D15		1.5
R24D16		1.6
R48D13	48	1.3
R48D14		1.4
R48D15		1.5
R48D16		1.6
R72D13	72	1.3
R72D14		1.4
R72D15		1.5
R72D16		1.6

the saturation process. More details of the saturation procedure were reported by Tang et al. (2020). This study focuses on the saturated hydraulic conductivity of a rooted soil with various dry densities and planting densities.

2.3. Test procedures of soil saturated hydraulic conductivity

After soil saturation, saturated hydraulic conductivity tests were performed following ASTM D-1586 (2008). Specimens were removed from the saturation container and placed in the permeameter before undergoing a variable head test in a permeameter. The experimental setup is depicted in Fig. 1. A water tank provided the continuous water supply for the whole seepage system. To prevent any residual air from affecting the saturated hydraulic conductivity of the specimens, it is essential to evacuate the air from the permeameter. After connecting the specimen to the permeameter and the water supply system, the inlet pipe is filled with de-aired water from the supply system. Once the water head stabilizes at approximately 1.7 m, de-aired water is introduced into the permeameter. Subsequently, the flatjaw pinchcock of air outlet is opened, and the permeameter is positioned at a tilted angle to allow for uninterrupted flow until overflow occurs without bubbles. Finally, it is essential to close the flatjaw pinchcock and level the permeameter prior to conducting formal tests. The saturated hydraulic conductivity was calculated for each specimen at nine different starting heads, using mean values for data analysis. The saturated hydraulic conductivity was calculated according to the formula as follows:

$$K_{sat} = 2.3\frac{aL}{At}log(\frac{H_1}{H_2})\# \quad (1)$$

where, K_{sat} (m/s) is the saturated hydraulic conductivity of soil, saturated hydraulic conductivity characterizes the conductivity of saturated porous media (Zeleke et al., 2005); a is the cross-sectional area of the variable head pipe (m^2); L is the height of the specimen (m); A is the cross-sectional area of the specimen (m^2); t (s) is the time elapsed for each saturated hydraulic conductivity test; H_1, H_2 (m) are the water head heights before and after the test, respectively, head denotes the kinetic energy associated with water flow (Youngs, 2000).

2.4. Measurement of root parameters

Once the K_{sat} of the tested rooted soil was determined, roots were carefully dug out from the specimens and soil adhering to the root was gently washed away, following the standardized root washing procedure adopted by Smucker et al. (1982). Then, the length of the roots was measured, and they were then placed in an oven for drying at 60 ± 1°C over a period of 72 h to obtain the root biomass (Ng et al., 2023a). Various root parameters can be derived by performing calculations using measurements of root length and biomass (Wu, 2017; Leung et al.,

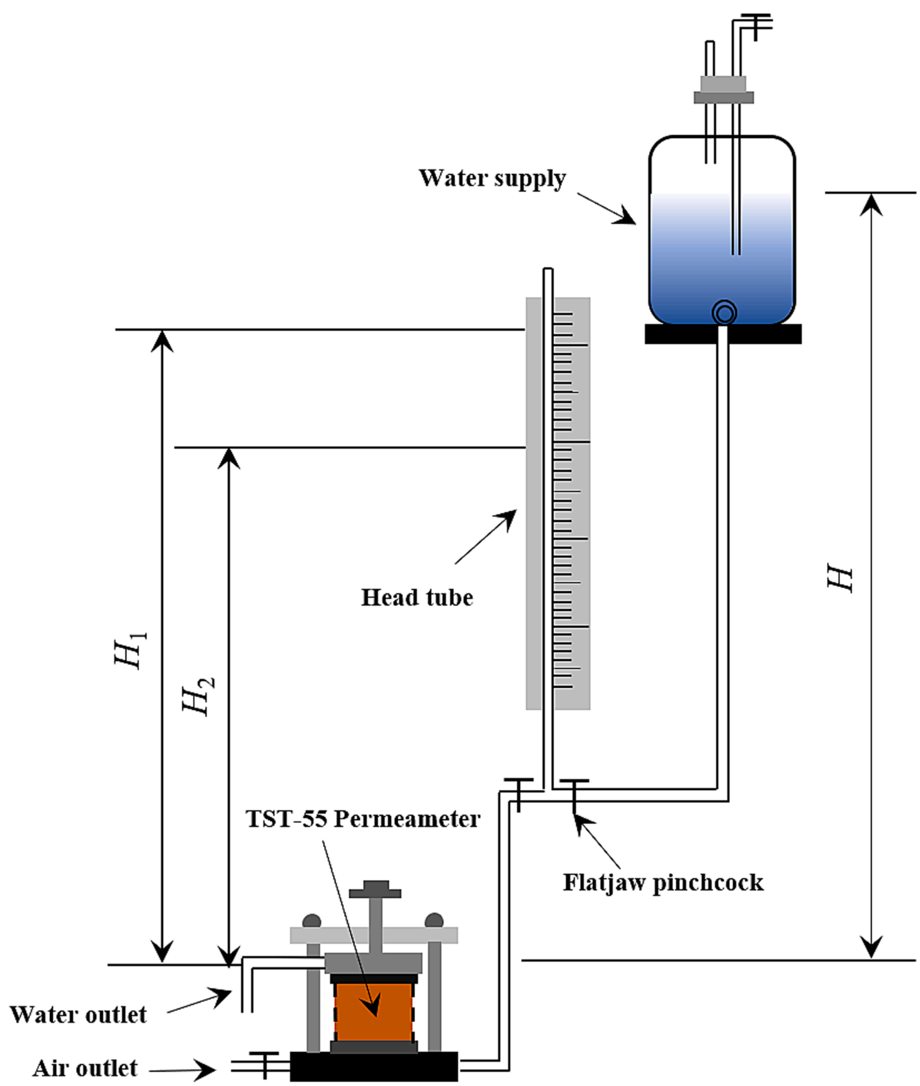

Fig.1. Schematic diagram of test device.

2018; Ng et al., 2020; Wang et al. 2023a, b):

(1) Root weight density $R_{w,d}$ (g/L):

$$R_{w,d} = \frac{R_b}{V_S} \# \quad (2)$$

where R_b is the root biomass (the weight of the roots after drying, g), it is quantified with an electronic analytical balance that has an accuracy of up to 0.0001 g; V_s is the volume of the cylindrical specimen (cm^3). Root weight density (g/L) indicates the total quantity of root organic matter per unit volume at a specific time. During plant root aging and death, root contractions form pores and roots transform into organic matter, aiding soil aggregate structure development. This is closely linked to root weight density: higher densities lead to larger pores and more organic matter following decay (Li et al., 2013).

(2) Root length density $R_{L,d}$ (cm/cm^3):

$$R_{L,d} = \frac{R_L}{V_s} \# \quad (3)$$

where R_L is the total length of roots in the cylindrical specimen (cm), it is measured using vernier calipers. Root length density (cm/cm^3) can effectively demonstrate the ability of roots to penetrate and interweave within the soil (Xu et al., 2011).

(3) Root length ratio $R_{L,r}$ (%):

$$R_{L,r(range)} = \frac{R_{L,range}}{R_L} \# \quad (4)$$

where $R_{L,r}$ is the root length ratio (%); Root length ratio (cm/cm) is the total length of roots within a certain length range. The content of long roots directly affects the range of nutrients available to plants. (Dotaniya and Meena, 2015) Therefore, the classification of roots into short, medium and long is of great value for assessing root growth. In this study, the majority of root lengths fall within the 0–4 cm range, with a few exceeding 4 cm. Hence, a root length of 0–2 cm is categorized as short, 2–4 cm as medium and >4 cm as long. $R_{L,r\ (S)}$, $R_{L,r\ (M)}$ and $R_{L,r\ (L)}$ represent root length ratio (0–2 cm), root length ratio (2–4 cm) and root length ratio (>4 cm), respectively.

2.5. Statistical analysis

One-way analysis of variance (ANOVA) along with the least significant difference (LSD) post hoc test was conducted to examine variations in soil dry density and planting density. The differences in $R_{w,d}$, $R_{L,d}$ and $R_{L,r}$ across various soil dry densities and planting densities were evaluated using the same statistic methods. The Pearson correlation coefficient was employed to quantify the relationship between the saturated hydraulic conductivity and root parameters. All significant differences were assessed at a significance level of 0.05.

3. Experimental results and discussion

3.1. Effects of dry density and planting density on root characteristics

The interaction between plants and soil results in changes in different soil dry densities and planting density conditions, which in turn affect root growth and development, thereby yielding distinct root parameters. Fig. 2 depicts the root weight density of rooted soil specimens with different dry densities and planting densities. For the rooted soil specimens with a given dry density, root weight density rises with higher planting densities. This is because higher planting density increases root biomass, leading to higher root weight density. Under constant planting densities, root weight density typically increases then decreases with increasing dry density. Poor root biomass in ryegrass with higher or lower dry density may be attributed to limited water and nutrient availability. When the soil dry density is high (1.5 and 1.6 Mg/m^3), roots can only access nutrients within a restricted area nearby, leading to a rapid depletion (Stirzaker et al., 1996). Futhermore, low soil dry density (1.3 Mg/m^3) can impair root biomass because there is less soil mass per volume (Cornish et al., 1984), or inadequate contact between roots and soil, hindering water and nutrient uptake (Herkelrath et al., 1977; Veen et al., 1992). Besides, soil with a high dry density (1.5 and 1.6 Mg/m^3) is often characterized by poor water drainage, which can lead to excessive moisture levels that adversely affect root respiration, inhibit root development and may ultimately result in root decay and mortality, thereby reducing overall root biomass (Raich and Tufekciogul, 2000;

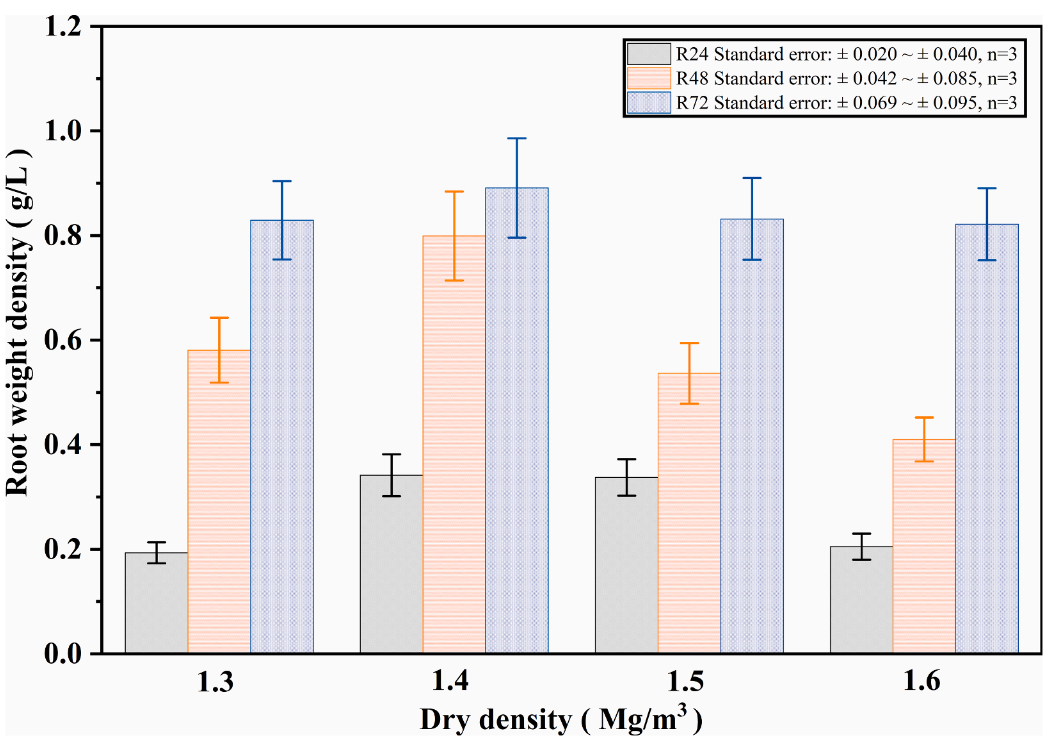

Fig.2. Effects of dry density and planting density on root weight density.

Rebetzke et al., 2014; Jin et al., 2015). Yet, excessively loose soil (1.3 Mg/m^3) can precipitate rapid loss of both water and nutrients, hindering plants' ability to access sufficient resources from the substrate and consequently diminishing root biomass (Herkelrath et al., 1977; Veen et al., 1992). Thus, optimal growth and development of plant roots necessitate an appropriate dry density within the soil. For the soil specimens involved in this study, the maximum root weight density occurs in the specimens with a dry density of 1.4 Mg/m^3. The peak values are approximately 0.34, 0.80, and 0.89 g/L, corresponding to each incremental increase in planting density. Therefore, this study concludes that the maximum ryegrass biomass was achieved at a dry density of 1.4 Mg/m^3.

In addition to root biomass, soil dry density and planting density also affect root length. Fig. 3 illustrates the influence of both dry density and planting density on root length density. As shown in the figure, an increase in soil dry density from 1.3 to 1.6 Mg/m^3 leads to a significant decrease in root length density for specimens with planting densities of 72 and 48 g/m^2, from 2.231 cm/cm^3 to 0.937 cm/cm^3 and from 1.85 cm/cm^3 to 0.725 cm/cm^3, respectively. The reduced root length density observed in denser soil can be attributed to increased mechanical resistance for the root when penetrating through the soil pore (Bengough and Mullins, 1990; Ng et al., 2014). Moreover, dense soil restricts air circulation, leading to a diminished oxygen content within the substrate, which adversely impacts root respiration and subsequently hinders the growth and development of roots (Pearson, 1966; Raich and Tufekciogul, 2000). In contrast with the high planting density (48 and 72 Mg/m^2), the specimens with a low planting density of 24 g/m^2 show an increasing root length density when the soil dry density increases from 1.3 to 1.4 Mg/m^3. After that, the root length density decreases with an increasing soil dry density. This may be because when the dry density increased from 1.3 to 1.4 Mg/m^3, the root biomass increased by 76.7 % at the planting density of 24 g/m^2, which is much larger than the 36.7 % and 7.4 % at the planting density of 48 and 72 g/m^2. Consequently, an increase in total root length is related to the substantial enhancement in root biomass, thereby contributing to a certain extent to the rise in root length density. Moreover, under a certain soil dry density, higher planting densities correspond to higher root length densities. As dry density increases, the influence of planting density on root length density gradually decreases. Specifically, at a dry density of 1.3 Mg/m^3, the root length density of a specimen with a planting density of 72 g/m^2 (2.23 cm/cm^3) is 3.4 times greater than that of a specimen with a planting density of 24 g/m^2 (0.66 cm/cm^3). At a dry density of 1.6 Mg/m^3, this difference decreases to 2.4 times. At a higher planting density, plant roots have greater demand and competition for water (Ng et al., 2016). The elevated soil water consumption results in a notable decline in root activity (Casper and Jackson, 1997; Jiang et al., 2013). These account for the greater reduction in root length density at a planting density of 72 compared to 24 g/m^2 as the dry density increases from 1.3 to 1.6 Mg/m^3.

Fig. 4(a) shows the effect of dry density on root length ratio for soil specimens with planting density of 24 g/m^2. An increase in dry density from 1.3 Mg/m^3 to 1.6 Mg/m^3 results in a rise in the root length ratio (0–2 cm) from 9.7 % to 44.0 %, while the root length ratio (>4 cm) drops from 48.7 % to 5.1 %. This is likely due to the denser arrangement of soil particles and the reduction in soil pore volume associated with larger dry density. This results in reduced space for normal root growth and heightened mechanical impedance between roots and soil particles (Jin et al., 2017). Consequently, it becomes difficult for roots to penetrate and grow through the soil, hindering the development of long roots (those longer than 4 cm in this study). In response, an increase in shorter roots occurs to absorb nutrients from the vicinity, resulting in a decline in total root length, root length density, and the proportion of long roots (>4 cm), alongside a simultaneous increase in the proportion of shorter roots (0–2 cm). A similar scenario is observed for specimens with higher planting densities, as depicted in Fig. 4(b) and (c). Furthermore, it can be inferred that at a soil dry density of 1.3 Mg/m^3, the ratio of root length (>4 cm) decreases from 48.7 % to 19.9 % as planting density increases from 24 to 72 g/m^2. However, under other dry density conditions, the change in root length ratio (>4 cm) is not significant. Regarding the ratio of root length (0–2 cm), as planting density increases from 24 to 72 g/m^2 and dry density increases from 1.3 to 1.6 g/m^3, the ratio remains approximately at 10 %, 15 %, 28 % and 43 %, respectively. As planting density increases, the distance between individual plants decreases, leading to heightened competition among roots for limited nutrients and living space. The reduced availability of area for nutrient absorption imposes "nutrient stress" on the plant (Goldberg and Miller, 1990), further inhibiting the development of long roots.

In summary, dry density and planting density significantly affect root

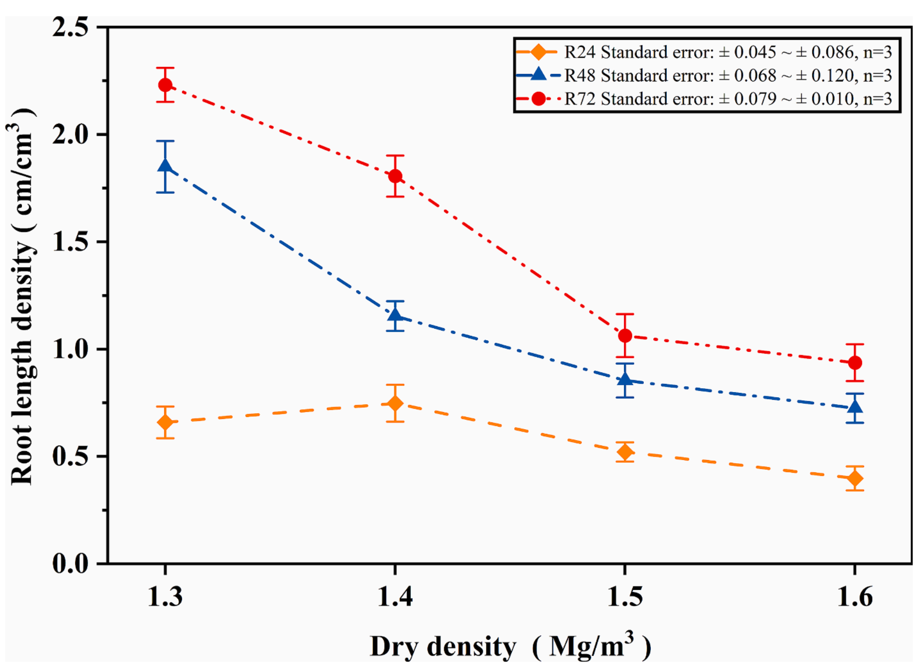

Fig.3. Effects of dry density and planting density on root length density.

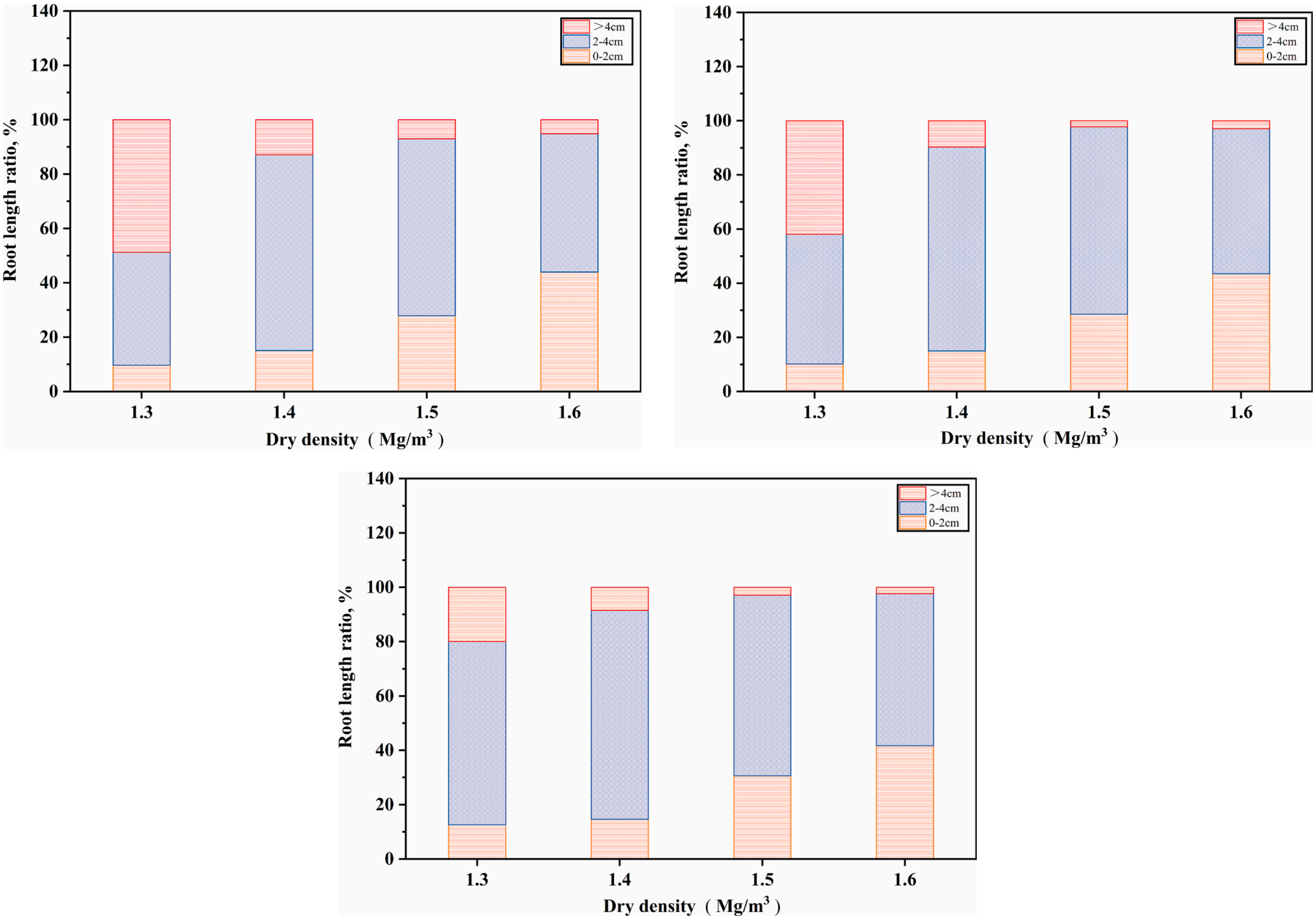

Fig.4. The root length ratio exhibits variations in response to dry density across different planting densities: (a) 24 g/m^2; (b) 48 g/m^2; (c) 72 g/m^2.

growth. The compactness of the soil determines the living space of the root and the difficulty of overcoming mechanical resistance, thereby directly impacting plant respiration. And planting density is closely associated with "nutrient and light competition" among plants. Root parameters serve as crucial indicators for evaluating plant root growth. Investigating the effects of dry density and planting density on root parameters establishes a foundation for subsequent research on the relationship between root parameters and saturated hydraulic

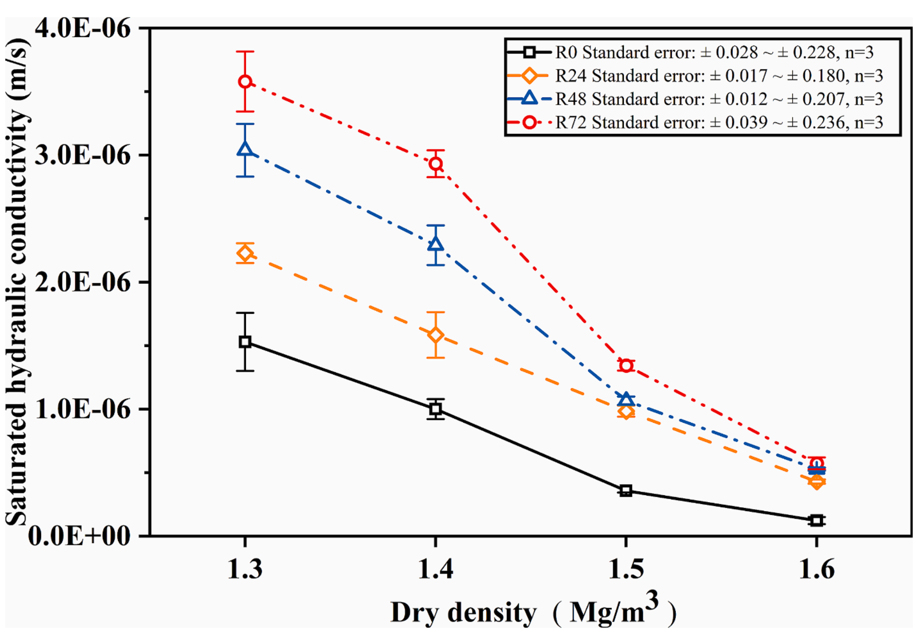

Fig.5. Saturated hydraulic conductivity of a rooted soil with different soil dry densities (1.3, 1.4, 1.5. 1.6 Mg/m^3) and planting densities (0, 24, 48, 72 g/m^2).

conductivity of rooted soil.

3.2. Saturated hydraulic conductivity of a rooted soil incorporating dry density and planting density

Fig. 5 illustrates the saturated hydraulic conductivity of soil specimens with various dry densities and planting densities. For the bare soil specimen, when the dry density increases from 1.3 to 1.6 Mg/m^3, the saturated hydraulic conductivity decreases from 1.53×10^{-6} m/s to 1.23×10^{-7} m/s, which is reduced by 92 %. This reduction is attributed to a decrease in pore volume, leading to the encounter of more convoluted pathways by water, which hinders rapid water movement and reduces the overall soil saturated hydraulic conductivity (Lohse and Dietrich, 2005; Dexter and Richard, 2009). As soil dry density increases, the saturated hydraulic conductivity of rooted soil specimens with different planting densities also shows a decreasing trend. As can be seen from the figure, the effect of dry density on saturated hydraulic conductivity becomes more significant with larger planting density.

Moreover, the presence of ryegrass roots in soil enhances the saturated hydraulic conductivity compared with bare soil. With a certain soil dry density, the saturated hydraulic conductivity increases with rising planting density. This is due to the increase in planting density leads to a corresponding increase in root biomass, resulting in more roots penetrating small pores and breaking up large aggregates (Materechera et al., 1992). This process generates preferential flow channels, indicating that the formation of root channels may serve as an important role for preferential flow (Bargués et al., 2014; Cheng et al., 2017; Lu et al., 2020; Mair et al., 2022). As the planting density increases, the number of these channels rises, thereby enhancing the soil saturated hydraulic conductivity.

The change ratio of soil saturated hydraulic conductivity is defined by dividing the saturated hydraulic conductivity of soil with vegetation at a specific planting density by that of bare soil at the same dry density. Fig. 6 shows the variation in the change ratio of saturated hydraulic conductivity with soil dry density under different planting densities. As can be seen from the figure that, at a certain planting density, the change ratio of soil saturated hydraulic conductivity increases with higher dry density. For example, at a planting density of 72 g/m^2, increasing the dry density from 1.3 to 1.6 Mg/m^3 results in an increase in the change ratio of soil saturated hydraulic conductivity from 134.0 % to 365.9 %. An increase in soil dry density leads to a reduction in the available space for plant root growth, an increase in mechanical resistance during root growth, and greater difficulty in nutrient acquisition (Rosolem et al., 2002; Ng et al., 2014). Consequently, this accelerates the aging and death of plant roots. As the roots age and die, they shrink and creating

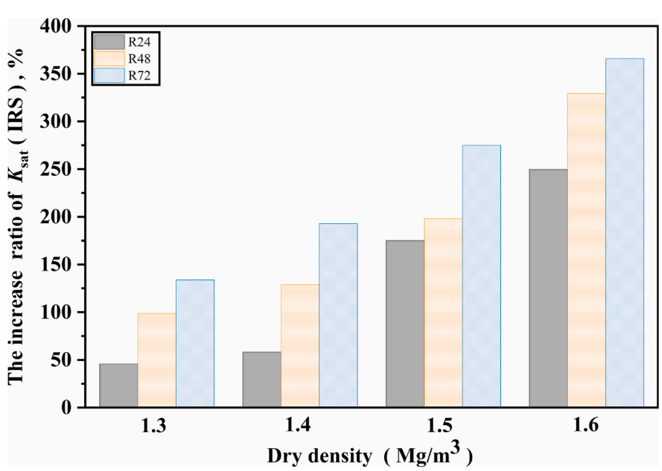

Fig.6. The increase ratio of soil saturated hydraulic conductivity compared with bare soil.

preferential flow channels at the root-soil interface, which significantly increases the saturated hydraulic conductivity of rooted soil (Ghestem et al., 2011; Ni et al., 2019).

The change ratios of soil saturated hydraulic conductivity in this study are all positive, indicating that the saturated hydraulic conductivity of the soil under grass vegetation is greater compared to the bare soil. Previous studies have demonstrated that the presence of plant roots in soil leads to a reduction in soil saturated hydraulic conductivity. Leung et al., 2015b observed that roots of *Schefflera heptaphylla* could decrease soil saturated hydraulic conductivity by approximately 50 %. The findings from Grayston et al. (1997) and Traore et al. (2000) are also consistent with this observation. This is mainly attributed to the fact that the plants used in their experiments were predominantly tree species, as opposed to a grass species utilized in this study. The larger root diameter of tree species results in more significant compression of surrounding soil particles during growth, leading to denser soil in the rhizosphere region and subsequently reducing soil saturated hydraulic conductivity (Aravena et al., 2011). While for the roots of grass species, the root diameter is smaller, resulting in less noticeable extrusion effects on the surrounding soil particles. Moreover, the roots of grass species lead to the generation of preferential flow channels at the root-soil interface, thereby significantly increasing soil saturated hydraulic conductivity (Banwart et al., 2019; Shi et al., 2021).

3.3. Relationship between root characteristics and soil saturated hydraulic conductivity

The correlation analysis and p-value test results are presented in Fig. 7 and Table 3, respectively. The soil saturated hydraulic conductivity has highly significant positive relationship with the root length density ($|r| = 0.82$, $p < 0.01$), as well as significant positive relationship with the root length ratio (>4 cm) ($|r| = 0.64$, $p < 0.01$). Moreover, there exists a moderately positive relationship for root weight density ($|r| = 0.54$, $p < 0.05$). Conversely, the relationship between root length ratio (0–2 cm) and root length ratio (2–4 cm) exhibits weak correlations ($|r| < 0.44$). The statistical analysis with multiple-regression highlights that the main root parameters that affects soil saturated hydraulic conductivity is root length density, followed by root length ratio (>4 cm) and root weight density.

Consequently, in order to elucidate more clearly the relationship between individual root parameters and saturated hydraulic conductivity of soil, three primary root parameters ($R_{w,d}$, $R_{L,d}$ and $R_{L,r}$ (>4)) were analyzed quantitatively and these analysis results are depicted in Fig. 8. The relationship between root weight density and soil saturated hydraulic conductivity is illustrated in Fig. 8 (a). As shown in the figure, an increase in root weight density from 0 to approximately 0.8 g/L leads to a significant decrease in soil saturated hydraulic conductivity for specimens with dry densities of 1.3 and 1.4 Mg/m^3, from 1.53×10^{-6} m/s to 3.58×10^{-6} m/s and from 1.00×10^{-6} m/s to 2.93×10^{-6} m/s, respectively. Specimens with higher dry densities of 1.5 and 1.6 Mg/m^3 also show an increasing trend in saturated hydraulic conductivity of soil, with increasing root weight density. These are mainly related to the increase in organic matter content in soil as root biomass expands. Increased root biomass enhances the capacity of roots to absorb water and nutrients from the soil, thereby overcoming the osmic of nutrients and promoting improved plant growth (Man et al., 2016; Ng et al., 2023b). During periods of rapid growth, plant roots secrete increased amounts of organic mucilage into the soil, which binds soil particles together and facilitates the formation of larger aggregate structures (Read et al., 2003; Galloway et al., 2018). Additionally, mucilage absorbs water and expands, resulting in alterations to pore structure (Carminati et al., 2016). Furthermore, mucilage may enhance microbial reproduction and metabolism within the soil, increasing microbial activity and leading to a greater abundance of microbial fragments and mycelium – both of which serve as binding agents that significantly contribute to the formation of large aggregates (Dennis et al., 2010; Chai

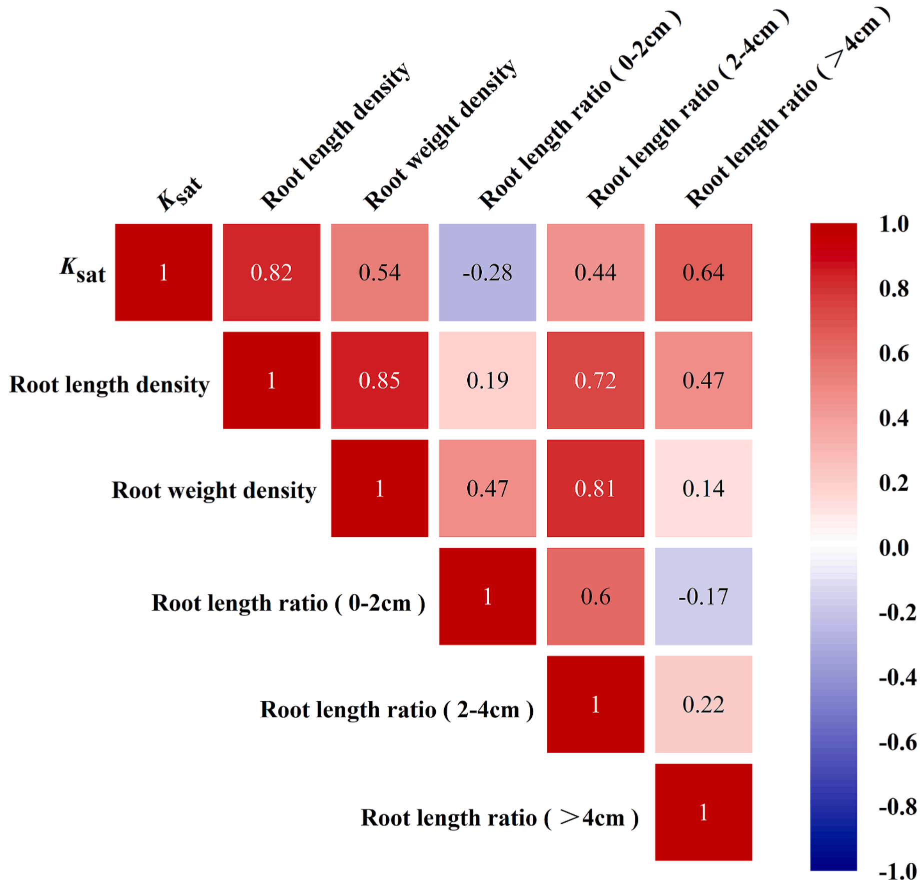

Fig.7. Pearson correlation coefficient analysis between root parameters and the saturated hydraulic conductivity of soil.

Table 3
Matrix showing correlations (correlation coefficient) between root parameters and saturated hydraulic conductivity of rooted soil.

Items	$R_{w,d}$	$R_{L,d}$	$R_{L,r\ (0\text{-}2)}$	$R_{L,r\ (2\text{-}4)}$	$R_{L,r\ (>4)}$
K_{sat}	0.54*	0.82**	−0.28	0.44	0.64**

Note: K_{sat}, the saturated hydraulic conductivity of soil; *Correlation is significant at the $p < 0.05$ probability level, **Correlation is significant at the $p < 0.01$ probability level, respectively.

and Schachtman, 2022). Therefore, as root weight density increases, root tips secrete greater amounts of organic mucilage. This enhancement in mucilage content facilitates the formation of larger aggregate structures within the soil, improves connectivity between aggregates, accelerates water flow, and consequently enhances the saturated hydraulic conductivity of soil (Galloway et al., 2018). It is important to note that an increase in root weight density also intensifies nutrient competition among roots, accelerates root senescence and mortality, resulting in numerous voids around roots that further augment the saturated hydraulic conductivity of the soil (Li et al., 2013).

Fig. 8 (b) demonstrates that soil saturated hydraulic conductivity increases with the rise in root length density. As can be seen from the figure that, at a certain dry density, soil saturated hydraulic conductivity increases with higher root length density. For example, at a dry density of 1.4 Mg/m^3, increasing the root length density from 0 to 1.81 cm/cm^3 results in an increase in soil saturated hydraulic conductivity from 1.00 to 2.93×10^{-6} m/s, soil saturated hydraulic conductivity increases by 193 %. This trend is likely due to the growth dynamics of ryegrass roots, which naturally penetrate and split soil structure, thereby facilitating the formation of pores between soil particles and enhancing both aeration and water conductivity (Lucas et al., 2019; Wang et al., 2020). Furthermore, root hairs play a crucial role in binding soil particles together, significantly increasing pore space among aggregates (Haling et al., 2013; Banwart et al., 2019). Additionally, roots absorb water from the soil during plant growth, as root length increases, so does the capacity for water absorption, leading to an increased number of voids within the rhizosphere (Materechera et al., 1992; Colombi et al., 2021). Root length density primarily influences the saturated hydraulic conductivity of soil through the mechanical actions (penetration and splitting) exerted by plant roots during their growth. As root length density increases, these mechanical effects intensify, enhancing the spatial arrangement and connectivity of rhizosphere soil pores, thereby augmenting saturated hydraulic conductivity (Lucas et al., 2019; Wang et al., 2020).

In addition, root length ratio also affects soil saturated hydraulic conductivity. Fig. 8(c) presents that an increased ratio of long roots (>4 cm) correlates with enhanced soil saturated hydraulic conductivity. Under constant planting densities, soil saturated hydraulic conductivity increases quickly and then increases slowly with increasing root length ratio (>4 cm). As shown in the figure, the higher the planting density, the more rapid the increase in soil saturated hydraulic conductivity. When the planting density was 72 and 24 g/m^2, the root length ratio (>4 cm) increased from 2.3 % to 19.9 % and from 5.1 % to 48.7 %, while the soil saturated hydraulic conductivity increased from 5.7×10^{-7} m/s to 3.58×10^{-6} m/s and from 2.3×10^{-7} m/s to 2.23×10^{-6} m/s, respectively. This phenomenon arises from the continuous extension and interweaving of elongated roots during their development, thereby creating preferential flow channels at the root-soil interface (Shao et al., 2017; Li et al., 2023). Research conducted by Reynolds (1966) and others has demonstrated that water traverses these channels with greater ease. Consequently, a higher proportion of long roots leads to more pronounced interweaving and additional water flow pathways,

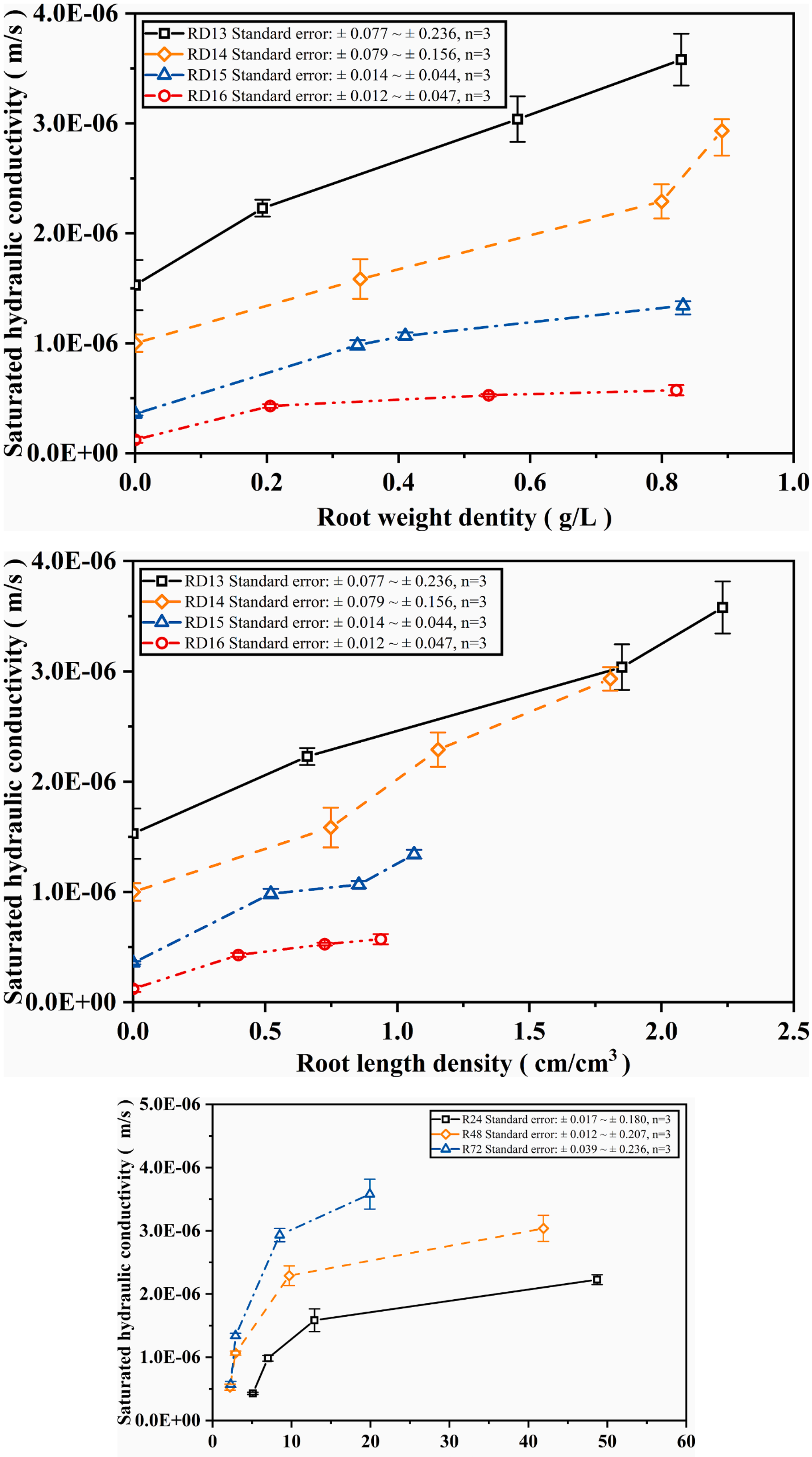

Fig.8. Relationship between root parameters and the saturated hydraulic conductivity of rooted soil: (a) Root weight density; (b) Root length density; (c) Root length ratio (>4cm).

thereby enhancing water transport and improving the saturated hydraulic conductivity of the soil.

According to this study, plant roots can significantly enhance the saturated hydraulic conductivity of soil. Specifically, the saturated hydraulic conductivity exhibits a linear positive correlation with root length density, root length ratio (>4 cm) and root weight density, with the strength of correlation diminishing in that order. Therefore, in complex field environments, sites can be sampled to select key root parameters for measurement. Subsequently, the saturated hydraulic conductivity of soil can be qualitatively assessed based on the established relationships between these root parameters and soil saturated hydraulic conductivity. This approach circumvents the limitations associated with time-consuming and labor-intensive direct measurements of soil saturated hydraulic conductivity. Furthermore, understanding the relationship between root parameters and soil saturated hydraulic conductivity can aid in optimizing agricultural irrigation system designs to ensure effective water infiltration into the soil and improve water use efficiency (Clothier and Green, 1994). Simultaneously, it provides a scientific foundation for designing urban drainage systems to facilitate rapid rainwater penetration into the ground, thereby mitigating risks associated with urban flooding and waterlogging (Kaur et al., 2020). Moreover, it can also provide guidance for engineering applications involving plants, such as green slope engineering and ecological restoration.

4. Conclusions

This study elucidates the relationship between root characteristics and the saturated hydraulic conductivity of grassed soil. The saturated hydraulic conductivity of soil specimens with different dry densities and planting densities was tested. Moreover, root parameters, including root length ratio, root length and root weight density, were measured. The stepwise multiple regression analysis of root parameters and soil saturated hydraulic conductivity were also conducted. The following conclusions can be drawn:

(1) As dry density increases, the mechanical resistance of soil particles to root growth leads to a decrease in both root length density and root length ratio of long roots (longer than 4 cm in this study), while the root length ratio of short roots (0–2 cm long in this study) increases. Root weight density initially increases and then decreases with increasing dry density, peaking at a dry density of 1.4 Mg/m^3, which is the optimal dry density for root biomass propagation.
(2) Root length density and root weight density both increased with higher planting density, whereas the root length ratio of long roots exhibited an opposite trend. Higher planting density results in more biomass but also intensifies nutrient competition, which is detrimental to sustained root growth.
(3) For both bare and grassed soil specimens, larger dry density results in smaller saturated hydraulic conductivity, yet the permeable effects become more significant with larger planting density.
(4) Grass roots increase the soil saturated hydraulic conductivity. With higher planting density, roots induce more preferential flow channels in the soil, leading to an increase in saturated hydraulic conductivity.
(5) Soil saturated hydraulic conductivity exhibits a positive correlation with root length density, root length ratio (>4 cm), and root weight density. Root length density exerts the most significant influence on soil saturated hydraulic conductivity ($|r| = 0.82$, $p < 0.01$), followed by root length ratio (>4 cm) and root weight density.

CRediT authorship contribution statement

Ling-Xin Cui: Writing – original draft, Methodology, Investigation. **Qing Cheng:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Pui San So:** Writing – review & editing. **Chao-Sheng Tang:** Writing – review & editing. **Ben-Gang Tian:** Methodology, Investigation. **Cong-Ying Li:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was gratefully supported by the National Natural Science Foundation of China (Grant No. 42172290, 42230710), Natural Science Foundation of Jiangsu Province (Grant No. BK20221250) and the Fundamental Research Funds for the Central Universities (Grant No. 14380224).

Data availability

Data will be made available on request.

References

Aravena, J.E., Berli, M., Ghezzehei, T.A., Tyler, S.W., 2011. Effects of root-induced compaction on rhizosphere hydraulic properties-X-ray microtomography imaging and numerical simulations. Environ. Sci. Tech. 45 (2), 425–431.
Astm d 1586,, 2008. Standard Test Method for Standard Penetration Test (SPT) and Split-Barrel Sampling of Soils. American Society for Testing and Materials, West Conshohocken, Pa.
Astm d-2487,, 2017. Standard practice for classification of soils for engineering purposes (Unified Soil Classification System). American Society of Testing and Materials, West Conshohocken. 44, 4897–4907.
Astm d698–12,, 2012. Standard test methods for laboratory compaction characteristics of soil using standard effort (12400 ft-lbf/ft^3 (600 kN-m/m^3)). ASTM International, West Conshohocken, PA.
Banwart, S.A., Nikolaidis, N.P., Zhu, Y.G., Peacock, C.L., Sparks, D.L., 2019. Soil functions: connecting earth's critical zone. Annu. Rev. Earth Planet. Sci. 47, 333–359.
Bargués Tobella, A., Reese, H., Almaw, A., Bayala, J., Malmer, A., Laudon, H., Ilstedt, U., 2014. The effect of trees on preferential flow and soil infiltrability in an agroforestry parkland in semiarid Burkina Faso. Water Resour. Res. 50 (4), 3342–3354.
Bengough, A.G., Mullins, C.E., 1990. Mechanical impedance to root growth: a review of experimental techniques and root growth responses. J. Soil Sci. 41 (3), 341–358.
Ben-Hur, M., Yolcu, G., Uysal, H., Lado, M., Paz, A., 2009. Soil structure changes: aggregate size and soil texture effects on hydraulic conductivity under different saline and sodic conditions. Soil Res. 47 (7), 688–696.
Bordoloi, S., Ni, J., Ng, C.W.W., 2020. Soil desiccation cracking and its characterization in vegetated soil: A perspective review. Sci. Total Environ. 729, 138760.
Carminati, A., Zarebanadkouki, M., Kroener, E., Ahmed, M.A., Holz, M., 2016. Biophysical rhizosphere processes affecting root water uptake. Ann. Bot. 118 (4), 561–571.
Casper, B.B., Jackson, R.B., 1997. Plant competition underground. Annu. Rev. Ecol. Syst. 28 (1), 545–570.
Chai, Y.N., Schachtman, D.P., 2022. Root exudates impact plant performance under abiotic stress. Trends Plant Sci. 27 (1), 80–91.
Chen, L., Sela, S., Svoray, T., Assouline, S., 2013. The role of soil-surface sealing, microtopography, and vegetation patches in rainfall-runoff processes in semiarid areas. Water Resour. Res. 49 (9), 5585–5599.
Chen, J., Wu, Z., Zhao, T., Yang, H., Long, Q., He, Y., 2021. Rotation crop root performance and its effect on soil hydraulic properties in a clayey Utisol. Soil Tillage Res. 213, 105136.
Cheng, Q., Tang, C.S., Xu, D., Zeng, H., Shi, B., 2021. Water infiltration in a cracked soil considering effect of drying-wetting cycles. J. Hydrol. 593, 125640.
Cheng, Q., Gu, Y.D., Tang, C.S., Zhang, X.Y., Shi, B., 2024. Desiccation cracking behaviour of a vegetated soil incorporating planting density. Can. Geotech. J. 61 (1), 165–173.
Cheng, Y., Ogden, F.L., Zhu, J., 2017. Earthworms and tree roots: A model study of the effect of preferential flow paths on runoff generation and groundwater recharge in steep, saprolitic, tropical lowland catchments. Water Resour. Res. 53 (7), 5400–5419.
Clothier, B.E., Green, S.R., 1994. Rootzone processes and the efficient use of irrigation water. Agric Water Manag 25 (1), 1–12.
Colombi, T., Torres, L.C., Walter, A., Keller, T., 2018. Feedbacks between soil penetration resistance, root architecture and water uptake limit water accessibility and crop growth–A vicious circle. Sci. Total Environ. 626, 1026–1035.

Colombi, T., Kirchgessner, N., Iseskog, D., Alexandersson, S., Larsbo, M., Keller, T., 2021. A time-lapse imaging platform for quantification of soil crack development due to simulated root water uptake. Soil Tillage Res. 205, 104769.
Cornish, P.S., So, H.B., McWilliam, J.R., 1984. Effects of soil bulk density and water regimen on root growth and uptake of phosphorus by ryegrass. Aust. J. Agr. Res. 35 (5), 631–644.
De Baets, S., Poesen, J., Reubens, B., Wemans, K., De Baerdemaeker, J., Muys, B., 2008. Root tensile strength and root distribution of typical Mediterranean plant species and their contribution to soil shear strength. Plant and Soil 305, 207–226.
Dennis, P.G., Miller, A.J., Hirsch, P.R., 2010. Are root exudates more important than other sources of rhizodeposits in structuring rhizosphere bacterial communities? FEMS Microbiol. Ecol. 72 (3), 313–327.
Dexter, A.R., Richard, G., 2009. The saturated hydraulic conductivity of soils with n-modal pore size distributions. Geoderma 154 (1–2), 76–85.
Ding, D., Zhao, Y., Feng, H., Peng, X., Si, B., 2016. Using the double-exponential water retention equation to determine how soil pore-size distribution is linked to soil texture. Soil Tillage Res. 156, 119–130.
Dotaniya, M.L., Meena, V.D., 2015. Rhizosphere effect on nutrient availability in soil and its uptake by plants: a review. Proceedings of the National Academy of Sciences, India Section b: Biological Sciences 85, 1–12.
Fan, L., Lehmann, P., Zheng, C., Or, D., 2022. Vegetation-promoted soil structure inhibits hydrologic landslide triggering and alters carbon fluxes. Geophys. Res. Lett. 49 (18), e2022GL100389.
Fragaszy, R.J., Santamarina, J.C., Amekudzi, A., Assimaki, D., Bachus, R., Burns, S.E., Tsouris, C., 2011. Sustainable development and energy geotechnology—Potential roles for geotechnical engineering. KSCE J. Civ. Eng. 15, 611–621.
Galloway, A.F., Pedersen, M.J., Merry, B., Marcus, S.E., Blacker, J., Benning, L.G., Knox, J.P., 2018. Xyloglucan is released by plants and promotes soil particle aggregation. New Phytol. 217 (3), 1128–1136.
Ghestem, M., Sidle, R.C., Stokes, A., 2011. The influence of plant root systems on subsurface flow: implications for slope stability. Bioscience 61 (11), 869–879.
Goldberg, D.E., Miller, T.E., 1990. Effects of different resource additions of species diversity in an annual plant community. Ecology 71 (1), 213–225.
Gonzalez-Ollauri, A., Mickovski, S.B., 2017. Hydrological effect of vegetation against rainfall-induced landslides. J. Hydrol. 549, 374–387.
Grayston, S.J., Vaughan, D., Jones, D., 1997. Rhizosphere carbon flow in trees, in comparison with annual plants: the importance of root exudation and its impact on microbial activity and nutrient availability. Appl. Soil Ecol. 5 (1), 29–56.
Guo, H.W., Wu, H.W., Zhang, Q., Chen, R., 2022. Effects of plant growth on water infiltration into three-layer landfill cover system. J. Eng. Geol. 30 (5), 1731–1743.
Haling, R.E., Brown, L.K., Bengough, A.G., Young, I.M., Hallett, P.D., White, P.J., George, T.S., 2013. Root hairs improve root penetration, root–soil contact, and phosphorus acquisition in soils of different strength. J. Exp. Bot. 64 (12), 3711–3721.
Herkelrath, W.N., Miller, E.E., Gardner, W.R., 1977. Water uptake by plants: II. The root contact model. Soil Sci. Soc. Am. J. 41 (6), 1039–1043.
Hwang, S.I., 2004. Effect of texture on the performance of soil particle-size distribution models. Geoderma 123 (3–4), 363–371.
Jiang, W., Wang, K., Wu, Q., Dong, S., Liu, P., Zhang, J., 2013. Effects of narrow plant spacing on root distribution and physiological nitrogen use efficiency in summer maize. The Crop Journal 1 (1), 77–83.
Jin, K., White, P.J., Whalley, W.R., Shen, J., Shi, L., 2017. Shaping an optimal soil by root–soil interaction. Trends Plant Sci. 22 (10), 823–829.
Jin, X., Zhang, L., Gu, J., Zhao, C., Tian, J., He, C., 2015. Modelling the impacts of spatial heterogeneity in soil hydraulic properties on hydrological process in the upper reach of the Heihe River in the Qilian Mountains. Northwest China. Hydrological Processes 29 (15), 3318–3327.
Kaur, G., Singh, G., Motavalli, P.P., Nelson, K.A., Orlowski, J.M., Golden, B.R., 2020. Impacts and management strategies for crop production in waterlogged or flooded soils: A review. Agron. J. 112 (3), 1475–1501.
Kim, J.H., Fourcaud, T., Jourdan, C., et al., 2017. Vegetation as a driver of temporal variations in slope stability: the impact of hydrological processes. Geophys. Res. Lett. 44 (10), 4897–4907.
Ledford, S.H., Zimmer, M., Payan, D., 2020. Anthropogenic and biophysical controls on low flow hydrology in the southeastern United States. Water Resour. Res. 56 (9), e2020WR027098.
A.K. Leung J. Liu Z. Jiang When Nature Meets Technology: AI-Informed Discovery of Soil-Water-Root Physical Interaction Vol. 382 2023 EDP Sciences p. 21001).
Leung, A.K., Garg, A., Ng, C.W.W., 2015a. Effects of plant roots on soil-water retention and induced suction in vegetated soil. Eng. Geol. 193, 183–197.
Leung, A.K., Garg, A., Coo, J.L., Ng, C.W.W., Hau, B.C.H., 2015b. Effects of the roots of Cynodon dactylon and Schefflera heptaphylla on water infiltration rate and soil hydraulic conductivity. Hydrol. Process. 29 (15), 3342–3354.
Leung, A.K., Boldrin, D., Liang, T., Wu, Z.Y., Kamchoom, V., Bengough, A.G., 2018. Plant age effects on soil infiltration rate during early plant establishment. Géotechnique 68 (7), 646–652.
Li, J., Cui, P., Yin, Y., 2023. Field Observation and Micro-mechanism of Roots-induced Preferential Flow by Infiltration Experiment and Phase-field Method. J. Hydrol. 129756.
Li, J.X., He, B.H., Chen, Y., 2013. Root features of typical herb plants for hillslope protection and their effects on soil infiltration. Acta Ecol. Sin. 33 (5), 1535–1547.
Liu, Y., Cui, Z., Huang, Z., López-Vicente, M., Wu, G.L., 2019. Influence of soil moisture and plant roots on the soil infiltration capacity at different stages in arid grasslands of China. Catena 182, 104147.
Lohse, K.A., Dietrich, W.E., 2005. Contrasting effects of soil development on hydrological properties and flow paths. Water Resour. Res. 41 (12).
Lu, J., Zhang, Q., Werner, A.D., Li, Y., Jiang, S., Tan, Z., 2020. Root-induced changes of soil hydraulic properties–A review. J. Hydrol. 589, 125203.
Lucas, M., Schlüter, S., Vogel, H.J., Vetterlein, D., 2019. Roots compact the surrounding soil depending on the structures they encounter. Sci. Rep. 9 (1), 16236.
Mair, A., Dupuy, L.X., Ptashnyk, M., 2022. Model for water infiltration in vegetated soil with preferential flow oriented by plant roots. Plant and Soil 478 (1–2), 709–729.
Man, J., Shi, Y., Yu, Z., Zhang, Y., 2016. Root growth, soil water variation, and grain yield response of winter wheat to supplemental irrigation. Plant Prod. Sci. 19 (2), 193–205.
Materechera, S.A., Dexter, A.R., Alston, A.M., 1992. Formation of aggregates by plant roots in homogenised soils. Plant and Soil 142, 69–79.
Ng, C.W.W., Woon, K.X., Leung, A.K., Chu, L.M., 2013. Experimental investigation of induced suction distribution in a grass-covered soil. Ecol. Eng. 52, 219–223.
Ng, C.W.W., Leung, A.K., Woon, K.X., 2014. Effects of soil density on grass-induced suction distributions in compacted soil subjected to rainfall. Can. Geotech. J. 51 (3), 311–321.
Ng, C.W.W., Ni, J.J., Leung, A.K., Zhou, C., Wang, Z.J., 2016. Effects of planting density on tree growth and induced soil suction. Géotechnique 66 (9), 711–724.
Ng, C.W.W., Ni, J.J., Leung, A.K., 2020. Effects of plant growth and spacing on soil hydrological changes: a field study. Géotechnique 70 (10), 867–881.
Ng, C.W.W., San So, P., Coo, J.L., Lau, S.Y., Wong, J.T.F., 2023a. Interactions between nutrient types and soil hydrological properties on yield and quality of Pinellia ternata, a medicinal plant. Ind. Crop. Prod. 195, 116423.
Ng, C.W.W., San So, P., Wong, J.T.F., Lau, S.Y., 2023b. Intercropping of Pinellia ternata (herbal plant) with Sedum alfredii (Cd-hyperaccumulator) to reduce soil cadmium (Cd) absorption and improve yield. Environ. Pollut. 318, 120930.
Ni, J.J., Leung, A.K., Ng, C.W.W., So, P.S., 2017. Investigation of plant growth and transpiration-induced matric suction under mixed grass–tree conditions. Can. Geotech. J. 54 (4), 561–573.
Ni, J.J., Leung, A.K., Ng, C.W.W., 2019. Modelling effects of root growth and decay on soil water retention and permeability. Can. Geotech. J. 56 (7), 1049–1055.
Pandey, R.N., Pandey, R.S., 1982. Estimation of effective infiltration rates in cracked soils. J. Agric. Sci. 99 (3), 659–660.
Pearson, R.W., 1966. Soil environment and root development. Plant Environment and Efficient Water Use 95–126.
Raich, J.W., Tufekciogul, A., 2000. Vegetation and soil respiration: correlations and controls. Biogeochemistry 48, 71–90.
Ran, Q., Wang, F., Li, P., Ye, S., Tang, H., Gao, J., 2018. Effect of rainfall moving direction on surface flow and soil erosion processes on slopes with sealing. J. Hydrol. 567, 478–488.
Read, D.B., Bengough, A.G., Gregory, P.J., Crawford, J.W., Robinson, D., Scrimgeour, C. M., Zhang, X., 2003. Plant roots release phospholipid surfactants that modify the physical and chemical properties of soil. New Phytol. 157 (2), 315–326.
Rebetzke, G.J., Kirkegaard, J.A., Watt, M., Richards, R.A., 2014. Genetically vigorous wheat genotypes maintain superior early growth in no-till soils. Plant and Soil 377, 127–144.
Reynolds, E.R.C., 1966. The percolation of rainwater through soil demonstrated by fluorescent dyes. J. Soil Sci. 17 (1), 127–132.
Rosolem, C.A., Foloni, J.S.S., Tiritan, C.S., 2002. Root growth and nutrient accumulation in cover crops as affected by soil compaction. Soil Tillage Res. 65 (1), 109–115.
Shao, W., Ni, J., Leung, A.K., Su, Y., Ng, C.W.W., 2017. Analysis of plant root–induced preferential flow and pore-water pressure variation by a dual-permeability model. Can. Geotech. J. 54 (11), 1537–1552.
Shi, X., Qin, T., Yan, D., Tian, F., Wang, H., 2021. A meta-analysis on effects of root development on soil hydraulic properties. Geoderma 403, 115363.
Smucker, A.J.M., McBurney, S.L., Srivastava, A.K., 1982. Quantitative separation of roots from compacted soil profiles by the hydropneumatic elutriation system 1. Agron. J. 74 (3), 500–503.
Stirzaker, R.J., Passioura, J.B., Wilms, Y., 1996. Soil structure and plant growth: impact of bulk density and biopores. Plant and Soil 185, 151–162.
Stokes, A., Atger, C., Bengough, A.G., Fourcaud, T., Sidle, R.C., 2009. Desirable plant root traits for protecting natural and engineered slopes against landslides. Plant and Soil 324, 1–30.
Tang, C.S., Cheng, Q., Leng, T., Shi, B., Zeng, H., Inyang, H.I., 2020. Effects of wetting-drying cycles and desiccation cracks on mechanical behavior of an unsaturated soil. Catena 194, 104721.
Tang, C.S., Zhu, C., Cheng, Q., Zeng, H., Xu, J.J., Tian, B.G., Shi, B., 2021. Desiccation cracking of soils: a review of investigation approaches, underlying mechanisms, and influencing factors. Earth Sci. Rev. 216, 103586.
Traoré, O., Groleau-Renaud, V., Plantureux, S., Tubeileh, A., Boeuf-Tremblay, V., 2000. Effect of root mucilage and modelled root exudates on soil structure. Eur. J. Soil Sci. 51 (4), 575–581.
Usharani, K.V., Roopashree, K.M., Naik, D., 2019. Role of soil physical, chemical and biological properties for soil health improvement and sustainable agriculture. Journal of Pharmacognosy and Phytochemistry 8 (5), 1256–1267.
Veen, B.W., Van Noordwijk, M., De Willigen, P., Boone, F.R., Kooistra, M.J., 1992. Root-soil contact of maize, as measured by a thin-section technique: III. effects on shoot growth, nitrate and water uptake efficiency. Plant and Soil 139, 131–138.
Wang, M.C., Huang, C.C., 1984. Soil compaction and permeability prediction models. J. Environ. Eng. 110 (6), 1063–1083.
Wang, B., Liu, J., Li, Z., Morreale, S.J., Schneider, R.L., Xu, D., Lin, X., 2023a. The contributions of root morphological characteristics and soil property to soil infiltration in a reseeded desert steppe. Catena 225, 107020.
Wang, X., Ma, C., Wang, Y., Wang, Y., Li, T., Dai, Z., Li, M., 2020. Effect of root architecture on rainfall threshold for slope stability: variabilities in saturated

hydraulic conductivity and strength of root-soil composite. Landslides 17, 1965–1977.
Wang, P., Su, X., Zhou, Z., Wang, N., Zhu, B., 2023b. Differential effects of soil texture and root traits on the spatial variability of soil infiltrability under natural revegetation in the Loess Plateau of China. Catena 220, 106693.
Wu, H.W., 2017. Atmosphere-vegetation-soil interaction: theory and mechanism. Chinese Journal of Geotechnical Engineering 39, 1–47.
Wu, H., Shi, A., Ni, W., Zhao, L., Cheng, Z., Zhong, Q., 2024. Numerical simulation on potential landslide–induced wave hazards by a novel hybrid method. Eng. Geol. 331, 107429.
Xu, S., Zeng, B., Lei, S., Su, X., 2011. Root features of several flooding-tolerant plants and their roles in enhancing anti-erodibility of the soil in three gorges reservoir region. Acta Pedol. Sin. 48 (1), 160–167.
Youngs, E.G., 2000. Hydraulic conductivity of saturated soils. In: Soil and Environmental Analysis. CRC Press, pp. 153–194.
Zeleke, T.B., Si, B.C., 2005. Scaling relationships between saturated hydraulic conductivity and soil physical properties. Soil Sci. Soc. Am. J. 69 (6), 1691–1702.
Zhang, K., Tang, C.S., Jiang, N.J., Pan, X.H., Liu, B., Wang, Y.J., Shi, B., 2023. Microbial-induced carbonate precipitation (MICP) technology: a review on the fundamentals and engineering applications. Environ. Earth Sci. 82 (9), 229.
Zhou, Z.C., Shangguan, Z.P., 2007. The effects of ryegrass roots and shoots on loess erosion under simulated rainfall. Catena 70 (3), 350–355.